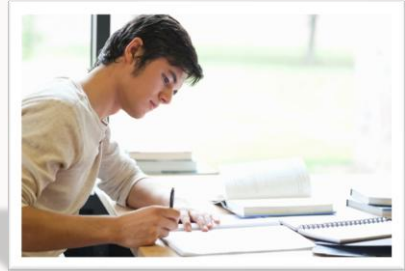




Database Foundations

2-1 Relational Databases



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Objectives

- This lesson covers the following objectives:
 - Describe the features of a single table
 - Describe the features and rules of a relational database
 - Describe the advantages and disadvantages of databases types
 - Define relational tables and key terms



Single Table

- As discussed previously, a flat file database is a type of database that stores data in a single table
- Flat file databases are generally in plain-text form, where each line holds only one record

RECORD	AUTHOR_ID	AUTHOR_NAME	TITLE
Record 1	AD0001	Oscar Wilde	A Vision
Record 2	AD0002	Leo Tolstoy	War and Peace
Record 3	AD0003	Oliver Goldsmith	Citizen of the World
Record 4	AD0003	Oliver Goldsmith	Androcles and the Lion



Advantages and Disadvantages of a Flat File Database

- Advantages

- Easy to understand
- Easy to implement
- Easy to extract information
- All records stored in one place
- Simple sorting and filtering of reports
- Less hardware and software requirements

- Disadvantages

- Less security
- Data inconsistency
- Data redundancy
- Cumbersome sharing of information
- Slow for huge databases



A flat file database is simpler and less expensive to use. It fits the needs of many small businesses and home users.

A flat file has many limitations because it contains ALL data in ONE table.

Here are some more disadvantages of flat file databases:

- No support for multi-user access
- Limited to smaller databases
- Limited functionality
- Integrity issues
- No security
- No backup and recovery
- No protection for data in concurrent access

Example of a Flat File Database

- Notice that both the details for the books and authors are stored in this single table, and there is a repetition of values

RECORD	AUTHOR_ID	AUTHOR_NAME	TITLE
Record 1	AD0001	Oscar Wilde	A Vision
Record 2	AD0002	Leo Tolstoy	War and Peace
Record 3	AD0003	Oliver Goldsmith	Citizen of the World
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Relational Databases

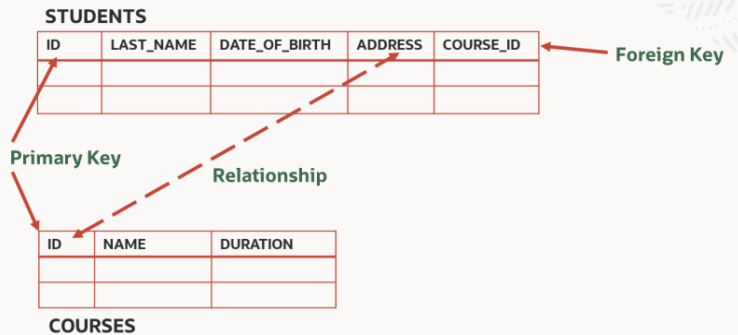
- A relational database presents information in tables with rows and columns
- Each column represents a particular type of information (a field), and each row lists one record
- The tables are then related to one another by using a common field
- A unique field called a key is used to identify each record in a relational database



A relational database management system (RDBMS) stores data in tables. Each table is given a name by the user who creates the table. The user generally chooses a name that correlates to the data that will be stored in the table; for example, STUDENTS, EMPLOYEES, LOCATIONS. When a table is created, the user also creates and names columns related to the specific characteristics that are stored for each record.

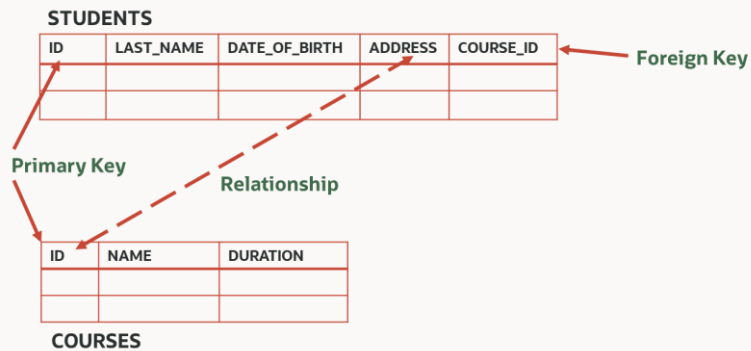
Relational Database: Example, 1 of 2

- Each table is assigned a PRIMARY_KEY column which uniquely identifies the entity instance
- A PRIMARY_KEY column in one table is designated as a FOREIGN_KEY column in a related table to form a relationship between the tables



Relational Database: Example, 2 of 2

- This relationship between the STUDENTS table and the COURSES table lets you store the data and query it to determine the specific courses that a student is attending (or has attended)



Advantages of a Relational (Multiple-Table) Database

- Less redundancy
- Avoidance of inconsistency
- Efficiency
- Data integrity
- Confidentiality

- *See Notes for more detail



Compared to the flat file system, the RDBMS has several advantages:

- **Less redundancy:** In a flat file system, there is much redundancy. For example, the author names are stored multiple times.
- **Avoidance of inconsistency :** If the same piece of information is stored in more than one place, then any changes in the data need to be made in all places where the data is stored.
- **Efficiency:** A database is usually more efficient than a flat file system, because a piece of information is stored in fewer locations.
- **Data integrity:** In a database system, it is easier to maintain data integrity because strong data types are assigned to each column.
- **Confidentiality:** It is easier to maintain the confidentiality of the information if data storage is centralized in one location.

Relational Tables

- A table is a simple structure where data is organized and stored

The diagram shows a table with 6 columns and 5 rows. Red arrows point from the word 'Columns' to each column header. Red arrows point from the word 'Rows' to each row of data. Below the table, three green arrows point to specific columns with labels: 'Primary Key Column (PK)' points to EMPLOYEE_ID, 'Foreign Key Column (FK)' points to DEPARTMENT_ID, and 'Unique Key Column (UK)' points to PAYROLL_ID.

EMPLOYEE_ID	LAST_NAME	FIRST_NAME	DEPARTMENT_ID	PAYROLL_ID	NICKNAME
100	SMITH	DANA	10	21215	Dana
310	ADAMS	TYLER	15	59877	Ty
210	CHEN	LAWRENCE	10	1101	Larry
405	GOMEZ	CARLOS	10	52	Chaz
378	LOUNGANI	NEIL	22	90386	Neil

Primary Key Column (PK) Foreign Key Column (FK) Unique Key Column (UK)

- Note: types of keys shown here will be discussed later in the lesson and course



DFo 2-1
Relational Databases

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The EMPLOYEE_ID is the Primary Key.

The PAYROLL_ID column is a unique key. This means that the system does not allow two rows with the same payroll_id.

In this example, the department_id foreign key refers to a row in the DEPARTMENTS table. You know that Dana Smith works in department 10. If you wanted to know more about Dana Smith's department, you would look for the department_id = 10 row in the DEPARTMENTS table.

Rules for Relational Database Tables

- Each table has a distinct name
- Each table may contain multiple rows
- Each table has a value to uniquely identify the rows
- Each column in a table has a unique name
- Entries in columns are single values
- Entries in columns are of the same kind
- Order of rows and columns is insignificant



Key Terms

- **Table** – A basic storage structure
- **Column** – attribute that describes the information in the table
- **Primary Key** – the unique identifier for each row
- **Foreign Key** – a column that refers to a primary key column in another table
- **Row** – data for one table instance
- **Field** – the one value found at the intersection of a row and column



Summary

- In this lesson, you should have learned how to:
 - Describe the features of a single table
 - Describe the features and rules of a relational database
 - Describe the advantages and disadvantages of databases types
 - Define relational tables and key terms



